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“He said it was being pierced all over his body. He was tired. He didn't want it anymore”: Care giver perceptions of blood glucose monitoring in children and adolescents living with type 1 diabetes Kyrgyzstan, Mali, Tanzania, and Peru

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Abstract

Background: Caregivers play an important role in the management of type 1 diabetes (T1DM) for children and adolescents. Controlling blood glucose levels is an essential part of diabetes management.
Previous studies have examined access to glucose monitoring devices and health system constraints in low- and middle-income countries and found that in many low- and middle-income countries (LMIC) tools for measuring blood glucose are unavailable and unaffordable. However, very little is known on the perception of caregivers in LMICs with regards to the important element of Self-Management of Blood Glucose (SMBG) and the emerging technologies and their suitability for these contexts.

Objective: The aim of this article will be to explore caregivers’ perceptions on how SMBG impacts to everyday life. For the purpose of this study the term caregiver is used referring to either a parent or an adult responsible for assisting with the management T1D with the child/adolescent

Methods: This qualitative descriptive study involved 24 interviews with a caregiver (either a parent or adults responsible for assisting with the management T1D with the child/adolescent) and a child or an adolescent living with T1DM identified through a mix of purposive and snowball sampling. Interviews were carried out based on common themes from a couple interview guide. This study was part of a larger study conducted in Kyrgyzstan, Mali, Peru and Tanzania. Interviews were analyzed using thematic analysis.

Results: The 24 couple interviews included 18 caregivers of minors living with T1D and 6 adolescents living with T1D. Across settings, caregivers described substantial psychological burden linked to repeated finger pricks, financial strain, and challenges managing diabetes during school and nighttime periods. The tools used raised issues for Blood Glucose Meters (BGM) of finger pricks, follow-up and affordability. Whereas for Continuous Glucose Monitors (CGM) issues of acceptability, accessibility and affordability of these devices was raised by the interviewees. Interviewees also highlighted different disruptions to daily life due to T1DM such as nutrition, school, and the need to access to medical care.

Conclusions: Caregivers’ perspectives on blood glucose monitoring highlighted the psychological and financial burden that weighs on them. Technological solutions, such as CGM, make it possible to alleviate certain obstacles of using a BGM and

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Background

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This qualitative descriptive study involved 24 interviews with a caregiver (either a parent or adults responsible for assisting with the management T1D with the child/adolescent) and a child or an adolescent living with T1DM identified through a mix of purposive and snowball sampling. Interviews were carried out based on common themes from a couple interview guide. This study was part of a larger study conducted in Kyrgyzstan, Mali, Peru and Tanzania. Interviews were analyzed using thematic analysis.

Results

The 24 couple interviews included 18 caregivers of minors living with T1D and 6 adolescents living with T1D. Interviewees highlighted different disruptions to daily life due to T1DM such as nutrition, school, and the need to access to medical care. Across settings, caregivers described substantial psychological burden linked to repeated finger pricks, financial strain, and challenges managing diabetes during school and nighttime periods. The tools used raised issues for Blood Glucose Meters (BGM) of finger pricks, follow-up and affordability. Whereas for Continuous Glucose Monitors (CGM) issues of acceptability, accessibility and affordability of these devices was raised by the interviewees.

Conclusions

Caregivers' perspectives on blood glucose monitoring highlighted the psychological and financial burden that weighs on them. Technological solutions, such as CGM, make it possible to alleviate certain obstacles of using a BGM and strips, but given their high cost can notably increase the financial burden.

1. Introduction

In 2017, there were an estimated 9 million people living with type 1 diabetes (T1DM), 27% in low- and middle-income countries (LMIC) [1]. Diabetes management focuses on maintaining blood glucose levels within a safe range to avoid acute hypoglycemia related to insulin therapy as well as long-term micro- and macrovascular complications correlated to the pathophysiology of chronic hyperglycemia [2, 3]. Diabetes complications represent an important determinant of lower quality of life, demands on the health system, economic costs, and increased morbidity and mortality [4, 5]. In order to manage blood glucose within daily insulin regimens, Self-Management of Blood Glucose (SMBG) in necessary. Different tools such as Blood Glucose Meters (BGM) and Continuous Glucose Monitors (CGM) help people with T1DM manage their insulin-doses and the body's reaction to food, physical activity, and glucose patterns and trends. BGMs and CGMs also provide a sense of security and confidence, which is beneficial for mental well-being. These tools also serve as a crucial tool for coordinating and sharing data with caregivers and healthcare professionals to enable effective treatment modifications and education [6]. This requires both availability and affordability of these tools as well as access to education and support from the health system which in many LMICs remains problematic [6-8]. A market report carried out in LMICs showed that there are still a number of factors that limit access to blood glucose monitoring devices, such as product characteristics, quality standards, cost and pricing, financing for product procurement, distribution and supply chain and provision of care [9].

Children and adolescents represent a specific population at risk of developing diabetes complications due to the complexity of diabetes management during this period. Current recommendations from the International Society for Pediatric and Adolescent Diabetes (ISPAD) indicate that glycated hemoglobin (HbA1c) targets for youth living with diabetes should be below 7% [10]. ISPAD also recommends a target of 4-10 mmol/L for blood glucose levels over time within the target range of CGMs [10]. For children and adolescents, parents or caregivers play an important role in care [11]. According to Markowitz et al. [11] the allocation of different diabetes-related tasks goes through an evolution as the child develops through early childhood, middle childhood/school-age years, and adolescence. In parallel to the changing roles the relationship between the parent/caregiver and child also changes with the needs, maturity and empowerment of the child and adolescent. These factors all place a certain level of stress on the parent/caregiver. It has been shown that psychosocial factors such as depression and family conflicts are common among young people with diabetes and can lead to suboptimal management of their diabetes [12, 13]. CGMs play a particularly valuable role in the management of diabetes in children and adolescents, providing notable benefits for the psychosocial well-being of young patients as well as their families [14]. However, these technologies are often unavailable and unaffordable in LMICs [15].

Previous studies have examined access to glucose monitoring devices and health system constraints in LMICs, however, very little is known on the perception of caregivers in LMICs with regards to the important element of SMBG and the emerging technologies such as CGM and their suitability for these contexts. Therefore, the aim of this article will be to explore caregivers' perceptions on how SMBG impacts to everyday life. For the purpose of this study the term caregiver is used referring to either a parent or an adult responsible for assisting with the management T1D with the child/adolescent.

2. Research design and methods

2.1. Study Design

This qualitative study is part of a larger study [16], which aimed to define requirements for non-invasive and minimally invasive glucose monitoring devices (NI-MI-GMD) in LMICs to inform a target product profile (TPP) to guide development and selection of suitable devices. Sampling, recruitment, training and data collection were carried out in Kyrgyzstan, Mali, Peru and Tanzania by research teams. The study protocol [16] was approved by Ethics Review Boards in each country. The overall study was sponsored by FIND (Foundation for Innovative Diagnostics, Geneva, Switzerland) and the Health Action International (Amsterdam) as part of the Addressing the Challenge and Constraints of Insulin Sources and Supply (ACCISS) Study.

2.2. Setting and participants

Participants were selected through a mix of purposive and snowball sampling through diabetes associations and other diabetes-related networks in Bishkek (Kyrgyzstan), Bamako (Mali), Lima (Peru), and Dar es Salaam (Tanzania). Table 1 provides the specifics of each country. This study focused on 24 couple interviews and 6 interviews per country with a caregiver and a minor living with T1DM. The interviews took place after obtaining informed consent from the caregiver and legal guardian of the minor.

2.3. Data collection

This qualitative study was carried out with couple interviews composed of a minor living with T1DM and his/her caregiver. Caregivers had to be the legal guardians of the minor. The interview guide consisted of a range of open-ended questions grouped in themes in order to respond to the objectives raised by the main study [16]. Some basic background information was collected from each interviewee, including gender of interviewee, gender of minor with T1DM and self-reported income level. In the interviews the following themes were discussed: diabetes, control of diabetes, the role of technology, glucose monitoring devices, requirements, societal, clinical, and technological factors, and cost. For some interviews, visual aids were used to help informants with insufficient knowledge of NI-MI-GMD to understand the type of technologies of interest in this study. The interviews were conducted in the local language, audio recorded and transcribed. Interviews from Kyrgyzstan (in Russian) and Tanzania (in Swahili) were translated into English. The interviews from Mali were in French and those from Peru in Spanish were not translated, but quotes used for the results were translated into English at a later stage for the purpose of this publication.

2.4. Analysis

All interviews were reviewed by the lead author, with annotations of relevant information. On the basis of these notes, the data was grouped in recurring themes and topics between interviewees and countries and to organize the information. Once an initial analysis was prepared the interviews were re-read and discussed between the lead author and DB. This process also enabled quotes to be retained according to their relevance and contributions based on the themes that emerged. Quotes were coded with a letter and a number to identify the respondent. The letter corresponds to the first letter of the country where the interview was carried out, and the number defines the different interviews from 1 to 6. The characteristics of study participants were grouped, summarized and

presented descriptively.
Data analysis has been carried out with care, rigor and a desire to bring the testimonies of interviewees to light in a transparent way.

3. Results

83.3% (20/24) of caregivers were women and 16.7% were men (4/24). The children and adolescents were aged between 4 and 18 years old. All interviews except for four were only done with a caregiver. These interviews included: a teenager living with diabetes (M5); a retired woman who looks after 11 minors with diabetes aged between 5 and 22 years old (T2); and two other interviews where two children living with T1DM were cared for jointly by the same caregiver (P4, T6).

Self-reported income status revealed that none of caregivers felt they belonged to a high-income household, 4% to a middle-high-income household (P2), 29% to a middle-income household (K1-5,M3,P1), 25% to a middle-low-income household (K6,T2,T3,M5,M6,P5), 33% to a low-income household (T1,T4,T5,T6,M4,P3,P4,P6) and 8% to a very low-income household (M1,M2).

Among the methods used to measure blood glucose levels, it was found that in Mali and Tanzania all used BGM, while in Kyrgyzstan everyone used a CGM, and some even used an insulin pump coupled with a CGM (K3, K4). In Peru, some used BGM (P3, P4) and others used CGMs (P1, P5, P6), and one used a CGM coupled to an insulin pump (P2).
The results will be presented according to the themes and sub-themes that emerged in relation to blood glucose monitoring, which are illustrated in Figure 1. The overall impact on everyday life of blood glucose management is presented, followed by the impacts of BGMs and CGMs to address some of these challenges.

3.1. Blood glucose meters

SMBG using a BGM device is still widely done with 58% (14/24) of all interviewees using this method. Note that even participants who use CGMs still have to use BGMs from time to time. The interviews highlight different challenges with regards to the use of BGMs, including the issue of finger pricks, the cost of testing, and follow-up.

3.1.1. Finger pricks

The finger prick is an integral part of the use of a BGM. All caregivers pointed out the problem that the children must prick themselves and underlined this with the fact that in children, this gesture could be easily associated with fear, screaming, crying, etc. They also mentioned the large number of pricks per day between blood glucose levels and insulin injections.

> “Sometimes they do not want to prick their hands because they are hard, calloused and, as they are thin, they have no muscle mass, so it hurts. Imagine the trauma of pricking their fingers to measure glucose, to take insulin, imagine the life they lead.” (P4, caregiver of a12-year-old child)

Moreover, some caregivers even shared their self-fears of pricking their own child.

> “The challenge comes for example when the patient condition worsens and for me I can’t prick her to read the test. I tell her sister to prick her because me I get scared,

[...].” (T1, caregiver of a 15-year-old adolescent)

However, the number of finger pricks for blood glucose measurement per day varied between the interviewees, and consequently the burden this represented. In Kyrgyzstan, caregivers spoke of an average of 7-10 pricks per day, while in Mali, Tanzania and Peru, it was more like 2-4 per day. M2 highlights the impact that 2 finger pricks a day can have on minors.

> “He also refused to check his levels twice a day. He said it was being pierced all over his body. He was tired. He didn't want it anymore.” (M2, caregiver of a16-year-old adolescent)

Despite this, the BGM is appreciated for its simplicity and the ease with which it can used by a child.

> “She is just testing herself, whenever she feels like she is not alright she tests.” (T4, caregiver of an 8-year-old child)

3.1.2. Follow-up

Blood glucose levels are a constant that varies throughout the day and night. One of the responsibilities of the caregivers is to be involved in this monitoring. However, more autonomous children prick themselves.

> “Often when he gets a problem it is from not observing the test results. He makes haste to inject then eat. That's where harm comes. So right now, I’m very careful, especially testing through the evening. I want when he measures, I see [the result].” (T6, caregiver of an unknown age child)

In addition, caregivers mentioned the necessity to accompany the child or teenager when blood glucose levels are measured at night, and K6 underline the consequent sleep issues.

> “Since my son got sick, […]. I go to bed late and wake up early. At night I have interrupted sleep. […]. At night when his glucose rises and he sleeps I inject insulin, then I observe how insulin acts.” (K6, caregiver of a 12-year-old child)

3.1.3. Affordability

With regards to the cost of BGMs, some caregivers mentioned donations for example from health facilities, diabetes associations or international programs, such as Life for a Child or Changing Diabetes in Children, covering part of their needs.

> “We didn't buy anything; they give us the devices and take care of everything. They give us the insulin, the injections and also the strips.” (M2, caregiver of a 16-year-old adolescent)

Sometimes families still had to buy extras or make up for the limited amount of donated equipment.

> “Yes, if you don't skip days just a few days the [strips] are over.” (T4, caregiver of a 9-year-old child)

However, most interviewees mentioned that there are many costs involved in managing diabetes.

> “Financial problems come first. Diabetes is not a disease when you have a normal financial situation.” (K3, caregiver of an 8-year-old child)

Different means of saving money are discussed by the interviewees, including for example reusing lancets as described by P3.

> “[...] we are not in a position right now to be spending lancets all the time because no, it is money, it is not possible.” (P3, caregiver of a 12-year-old child)

3.2. Continuous glucose monitors

The following results highlight the use of CGMs by the interviewees using this technology.

3.2.1. Acceptability

The use of CGMs allows for fewer finger pricking, blood glucose monitoring at shorter intervals, at a distance and during the night, and better control of glycated hemoglobin, as underlined below by K2.

> “Makes life easier because every time you monitor your child, running after the child to pierce her finger and check the glucose is a big problem for us. Before the sensor, the problems for us were to check her glucose at night. I had to wake her up, she would cry, and the mood would change for the worse. Now it’s much better with the sensor. Thanks to the sensor, our level of glycated glucose decreased from 11 to 9. This is because we quickly bring glucose to normal.” (K2, caregiver of a 4-year-old child)

The reliability of the device is essential for caregivers. T3 said that he would adapt his use of the device according to his knowledge of his child's condition.

> “(Laughing) I would trust them [new devices] a bit... I will look also on the child’s condition since I have experience, I can now know that now it [blood glucose] has dropped and now it has risen. So, I will trust it depending on my experience of taking care of him.” (T3, caregiver of an 18-year-old adolescent)

Caregivers underlined the problem that CGMs do not completely prevent finger pricks. P2 underlines the fact that some CGM devices require regular calibrations, and thus the use of finger pricks and the use of a glucose meter.

> “Yes, at least twice a day because my daughter uses an insulin pump, and her pump, the sensor that her pump uses requires two calibrations a day, mandatory.” (P2, caregiver of a 9-year-old child)

The length of use before changing the device is a crucial point for caregivers, who count their expenses monthly, and a 14-days device is recognized as being too short.

> “[...] you know what depends there are two things, although my financial capacity is limited, but it also depends on how long the device can last?” (T6, caregiver of an unknown aged child).

In addition, CGMs will have to be adapted to the cultural contexts and activities of the minors, as highlighted below by M6 and M5.

> “Yes, [discretion] would be very important, because illness is something, even when people know you have such and such an illness, it will complicate your marriage.” (M6, caregiver of a 15-year-old adolescent)

> “Something that's not a part of your body, God didn't create you like that, it's not a shirt that you wear, except something that's permanently stuck to your body and that marries with the blood, I think that can often bother sick people.” (M5, caregiver of a 17-year-old adolescent)

3.2.2. Accessibility

Most caregivers (except K6, P4) owned a smartphone and therefore had the potential access to the technology needed for CGM devices. A second condition for access depended on the caregivers’ knowledge of these new technologies. In Mali and Tanzania, none of the caregivers had heard of any device other than BGMs for measuring blood glucose levels. Another prerequisite is the skills needed to use it correctly, and caregivers required therapeutic education, as said by P4.

> “[…] it's not just to say "look, here, I'll give you a glucose sensor and that's it, put it on", I mean, first educate me, right? I mean, teach me, explain to me how it works, in other words, what use I can get out of the device, right, in other words, to make 100% use of it, not just, not just not pricking, not just putting it there to watch all day, in other words, what's the point of seeing a horrible graph above 200, 300 and not knowing how to improve it.” (P4, caregiver of an 11-year-old child)

There is also the problem of accessibility on the market and the sale of devices in pharmacies.

> “Ah... Well, it would have to be coordinated between them, between the Ministry of Health and the people who manufacture the products, so that the products can be delivered, and people can buy them.” (P6, caregiver of a 16-year-old adolescent)

3.2.3. Affordability

CGM devices represent a cost for the families who use them. To save money, some people buy the devices abroad. K3 underlined competitive prices abroad.

> “It would be nice if the sensor is sold here at the price that is sold in Russia. Now our company [name of company] sells a sensor for 6300 soms [approx. US$ 70]. In Russia, we buy [name of sensor] for 4300 rubles [approx. US$ 46]. You can order it for 3700 soms [approx. US$ 41] from France.” (K3, caregiver of an 8-years-old child)

Others extend the use of devices as mentioned by K4.

> “In general, you need to replace the sensor every 14 days, but our mothers have found a way to extend the life of this sensor, so of course we extend it. This saves our money.” (K4, caregiver of a 9-year-old child)

However, K1 mentioned the savings they make using CGM on other materials used for BGM.

> “Before the sensor, we spent 4,000 soms [approx. US$ 45] for test strips. Now when we installed the sensor, we buy test strips for 2,000 soms [approx. US$ 22].” (K1, caregiver of a 12-year-old child)

Globally for those who never used CGMs, the main issue was to ensure than any new technology represented less or the same expense than what they use. Realizing the significant costs involved, caregivers are prepared to pay. T4 spoke about the possibility to organize themselves and involve family members to raise the money to afford such devices.

> “We will sit together and see how it’s sold... maybe and the siblings will contribute money for purchasing it.” (T4, caregiver of a 9-year-old child)

A summary of the results related to BGMs and CGMs is presented in Table 2.

3.3. Impact on everyday life

Beyond the actual tools the management of type 1 diabetes has wide ranging impacts on these families. SMBG is impacted by and impacts such aspects as daily life activities, nutrition, school and access to medical care. Glycemic control involves adaptations in daily life for children, teenagers, and caregivers. K6 highlighted the impact of the condition on their lives.

> “Diabetes is an incurable disease. Living with diabetes is to be 24 hours in tension. There are no rest and holidays with this illness. For me personally this disease was a huge stress. My life has fully changed, and my relationship with my son has become also different as he is now in his awkward age. I am alone and it is difficult for me morally and financially.” (K6, caregiver of a 12-year-old child)

K4 emphasized the role of the caregiver and the stress experienced.

> “She is a child, she does not understand, but for parents, this is a big burden and stress.” (K4, caregiver of a 9-year-old child)

P3 and T6 also share the need to supervise the reading of blood glucose results.

> “[...] in fact, it was more of a challenge for me than anything else, it taught me that I have to be aware of my child, that I have to supervise him constantly [...].” (P3, caregiver of a 12-year-old child)

Overall, the main issues impacting daily management of T1DM, mentioned by caregivers were nutrition, school, and access to medical care.

3.3.1. Nutrition

Caregivers encountered difficulties with imposed dietary restrictions linked to glycemic control and the child's desires. K3 spoke about limiting the quantities of food and its impact on the child and the parent's role.

> “For mothers, it is psychological stress. It is also difficult to restrain the child in eating. She wants everything, and sometimes we can’t refuse, then hyperglycemia happens. Everything depends on us. You also need to learn how to calculate properly bread units.” (K3, caregiver of an 8-year-old child)

Moreover, the diet involves efforts in terms of cooking, habits, and costs as underlined below by K1,

P3 and T4.

> “Food restrictions is our problem. Another nuance is that expenses have increased, and for food too. Now we need to maintain proper nutrition, but it turns out to be much more expensive. The sugar substitute costs 5-6 times more than the sugar itself.” (K1, caregiver of a 12-year-old child)

> “And that's not all because it also depends on the quality of the food because it's not... it's not a food, for example, of eating rice with eggs, you have to give it fibers, you have to give it vitamins.” (P3, caregiver of a 12-year-old child)

> “The problems I get are on diet. On diet some people.... since I’m usually at work so that time of... because he has to eat four meals. So time for preparing him food becomes a problem and sometimes he doesn’t get those four meals. He can get three meals until I have gone extra miles to force him learn to cook on his own.” (T3, caregiver of an 18-year-old adolescent)

Some parents mention the need to keep an eye on the blood glucose results so as not to miss out on any dietary “misbehavior” on the part of the child.

> “[...] for me whenever I doubt that there is a possibility that she might lie to me, I tell her after you finish to test then show me the reading from the machine.” (T1, caregiver of a 15-year-old adolescent)

3.3.2.School

During the day, SMBG of non-autonomous young children who go to school can be a real challenge for caregivers. K2 highlighted the adaptation of the professional activity to be closer to the child.

> “[The mother] specifically applied for [a] job in [a] kindergarten to keep track of [her] child.” (K2, caregiver of a 4-year-old child)

K3 underlined the need of child’s autonomy for BGM.

> “Since September this year, when she went to school, my daughter began to use a blood glucose meter on her own. It was time for her to learn because I can’t be with her all the time.” (K3, caregiver of an 8-year-old child)

And for some children, blood glucose levels are not monitored during the school day, as M4 mentioned.

> “They're children, and I don't even allow her to bring the [BGM] to school because when they're playing with the other children, they might break it!” (M4, caregiver of a 17-year-old adolescent)

Diabetes-related complications can arise at any time, and T4 mentioned the difficulties he is experiencing with his child's school.

> “[…] They are scared because even now when she goes to school when she just feels bad, they tell me to go and take the child as she feels bad.” (T4, caregiver of 9-year-old child)

Extracurricular activities such as sport, which are part of a child's development, create another challenge for blood glucose management.

> “Before sport, blood sugar levels are good. But after sport, whoosh, he goes into hypoglycemia. I really don't know how to deal with that.” (M1, caregiver of a 15-year-old adolescent)

Not all caregivers felt that diabetes management was particularly restrictive in terms of the respective activities of the child and teenager, who can still lead a normal life according to them.

> “No, it hasn't made any difference to her work or school life, she never misses a day unless we have an appointment. Apart from that, there's no change in her daily life.” (M1, caregiver of a 15-year-old adolescent)

3.3.3. Access to medical care

Managing diabetes involves monthly medical consultations in specialist centers and access to emergency services. Caregivers point out the geographical and financial factors of transport involved. M2 said they received money from hospitals for transportation which is provided by Santé Diabète through support from Life for a Child. In addition, medical care involves additional costs which depend on health systems and differ between countries.

> “The hospital only gives 1000 CFA (US$ 1.66) to the patient. […] Otherwise, you've seen the distance yourself. So, you have to take a lot of fuel to get there and back, every time there's an appointment.” (M2, caregiver of a 16-year-old adolescent)

Proximity to care facilities seems to be important, whereas P3 said that they live closer to the hospital since his child was hospitalized, and T5 said that they chose the school according to its distance from a care center.

> “I live in Talara, I was being treated at the Talara facility, but my child required hospitalization because, I mean, he was in a delicate condition. He suffered a decompensation, and they sent me to Sullana, so until now I have been in the support hospital in Sullana and, what is it called, and well, I am still here because I am afraid that my child will deteriorate again. I am still here in Sullana.” (P3, caregiver of a 12-year-old child)

> “I sent him to these regular schools because outside the school there is Twangoma hospital.” (T5, caregiver of a 13-year-old adolescent)

4. Discussion

The concerns mentioned about blood glucose monitoring in children and adolescents living with T1DM varied from one caregiver to another, but mainly related to: repeated finger pricks; financial burden as a barrier to access to blood glucose monitoring devices; and psychological distress related to the management of diabetes, concerns about their child's health and blood glucose control, as well as the medical and medication costs incurred by the condition. This in turn had wider impacts on diet, school and access to medical care and the costs involved. Looking at the factors that seemed to alleviate the burden on caregivers, there was familiarity with T1D, including acquired knowledge and recognition of hypoglycemia symptoms, the child's increased autonomy, support from familiar surroundings, as well as the use of CGMs for blood glucose management.

In terms of the child's autonomy, CGMs appeared to play a valuable role in supporting this aspect. They helped the child better understand their hypoglycemia symptoms, continue extracurricular activities, reduce the need for frequent finger pricks, and more effectively manage their food intake and insulin dosage. CGM users appeared to be highly satisfied with the support these devices provided in their daily lives. They particularly highlighted the reduction in finger pricks, continuous monitoring, including overnight, valuable assistance in management of T1D, and the resulting alleviation of the psychological burden.

However, CGMs are more expensive than BGMs, which already seem to be financially difficult to access. Differences in CGM-use across countries likely reflects broader structural inequities in procurement systems, insurance coverage, and supply chains, as previously documented in LMIC health system assessments [8]. These disparities suggest that technological innovation alone will not reduce caregiver burden without parallel financing and policy reform. Device availability differed substantially across the four countries, reflecting variations in health system financing and market access. CGM use was universal among participants in Kyrgyzstan, whereas none of the caregivers in Mali or Tanzania reported access to CGM technology. In order to implement CGMs in the management of type 1 diabetes in children and adolescents, the focus must then be on their accessibility (market availability, medical knowledge, therapeutic education), their affordability as well as their acceptability, feasibility, and sustainability.

By definition, parents already have the role of caring for their children, but in the case of a child living with a chronic condition, the parent takes on a very special role, as the child's life depends on it [19]. The optimal management of blood glucose levels in diabetes is multidimensional and particularly demanding requiring major adaptations to daily life. The effective management of these different measures in children and adolescents is a major challenge for the parents concerned, as they have control over many of these elements, such as food, supplies, medication, appointments management, sports and school enrolment and transports. Existing literature in South Africa has also shown difficulties in term of caring of children with diabetes, such as emotional, social, practical and financial difficulties [20]. Raina et al. [21] have shown that higher caregiving demands were associated with lower psychological and physical health among caregivers of children with chronic conditions. Several determinants of parents, such as poor mental health, uncertainty about illness, poor self-efficacy, can negatively impact their ability to monitor their child's glycemic levels effectively, ultimately leading to suboptimal diabetes management in children [22, 23]. A qualitative study conducted in China highlighted some of the challenges faced by caregivers of children newly diagnosed with T1DM, such as psychological stress, difficulty family function’s changes, daily management, financial burden and the lack of social support for parental caregivers [24]. Markowitz and al. [11] have shown that the caregivers play an important role in supporting the adolescent in self-care and treatment adherence.

However, Wysocki and al. [25] showed that children with little or no parental supervision had poorer glycaemia control as a result of excessive self-care. CGMs have been touted as a means to support better blood glucose management. A study on the role of CGMs carried out in Pakistan among a population of people living with diabetes and mostly (90%) using BGMs found that a majority believe in the role that technology can play in supporting management of diabetes, by improving care, reducing complications, and simplifying management [26]. The study also put in evidence the affordability problem which seems to be a major barrier due to high out-of-pocket health related expenses. Burckhardt and al. [27] were able to show that use of remote monitoring with CGM significantly improved the daily experience of caregivers and the feeling of a better understanding of diabetes care and control. CGMs are among the emerging technologies designed to improve diabetes

management, particularly by enabling better glycemic control and reducing glucose variability. They are now widely used in high-income countries and have demonstrated positive outcomes, especially among children and adolescents, including improved time-in-range and reduced hypoglycemia [28].

This qualitative study may have some limitations in terms of data collection and sampling, as the interviews were conducted by different people in different countries and the couples interviewed were selected by the respective organizations in each country. Moreover, there may have biases in analysis and results presentation linked to the researchers based on their different knowledge and perspectives on the topic. Finally, the analysis was conducted by a researcher with little knowledge of the local cultures in three of the four countries, as well as the lived experience of type 1 diabetes. This may have resulted in contextual bias in analyzing and presenting the data.

To our knowledge this is the first study to explore the perceptions of caregivers in LMICs with regards to blood glucose management and the impact of the tools to manage this. Blood glucose management is a cornerstone of diabetes management to prevent morbidity and mortality. SMBG and the tools for this are essential. As highlighted in this article many barriers exist with regards to both the daily burden of SMBG as well as the cultural, technological and cost related aspects of the tools needed to do this. With the first ever global diabetes coverage targets including, “80% of people with diagnosed diabetes have good control of glycaemia” and “100% of people with type 1 diabetes have access to affordable insulin and blood glucose self-monitoring” [29] addressing all these barriers to proper SMBG are necessary.

This study is part of a larger study to define a TPP with the specificities for a new blood glucose measurement tools. Based on the findings of this study, the following recommendations outline the key characteristics that such a device should possess. First and foremost, the device should be socially accepted. Caregivers highlighted the importance of reliability, particularly in terms of accurate readings, as well as portability for children and adaptability to their school and extracurricular activities. It should also be durable, unbreakable, and reusable. Additionally, it must be user-friendly for children and supported by therapeutic education to ensure families use it properly. Moreover, the device should be easily accessible on the market and affordable, ideally at a lower cost than current options.

5. Conclusion

Self-monitoring of blood glucose seems to be a real challenge for caregivers. While BGMs are well implemented, the interviewees in this study mention several negative aspects to their day-to-day use, such as finger pricks, the physical presence of caregivers for monitoring and costs. CGM can be a real help in the management of diabetes, particularly in the case of caregivers looking after children and adolescents living with T1DM, by improving the empowerment mechanism for children, allowing them to better understand their condition and manage their diabetes more independently. Caregivers from LMICs would appear to be open to use CGMs to manage the diabetes of children and adolescents. However, there are concerns related to affordability and other factors related to low and middle-income settings that need to be considerate before implementing such devices in these contexts. Finally, device characteristics need to respect reliability, durability and context specific acceptability to children, adolescents, and their families to guarantee their use and justify their purchase. Beyond the tool a variety of other factors impacting the daily life of the person with T1D and their caregivers need to be addressed to ensure optimal use of new technologies.

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Conflicts of Interest
All authors report no conflicts of interest with regards to this work.

Data Availability
In safeguarding the anonymity of study informants in this qualitative study, and in respecting their right to safety, confidentiality and privacy, the transcripts of qualitative data collection encounters used in this study cannot be made available. The full dataset is available upon reasonable request from FIND.

Authors contributions
SP analyzed the data with support from DB. DB, ES, CH, ML, PS and BV developed the methodology and were responsible for the overall management and coordination of project as well as obtaining funding. AA, SB, AD, MLP, MN, SLP and KR provided critical input into the development of the methods and were responsible for in-country data collection and project management. All authors provided feedback and agreed on content of final submission.

Abbreviations
BGM – Blood Glucose Meter
CGM – Continuous Glucose Monitor
ISPAD – International Society of Pediatric and Adolescent Diabetes
LMIC – Low- and Middle-Income Country
NI-MI-GMD - non-invasive and minimally invasive glucose monitoring devices
T1DM – type 1 diabetes
TPP – Target Product Profile

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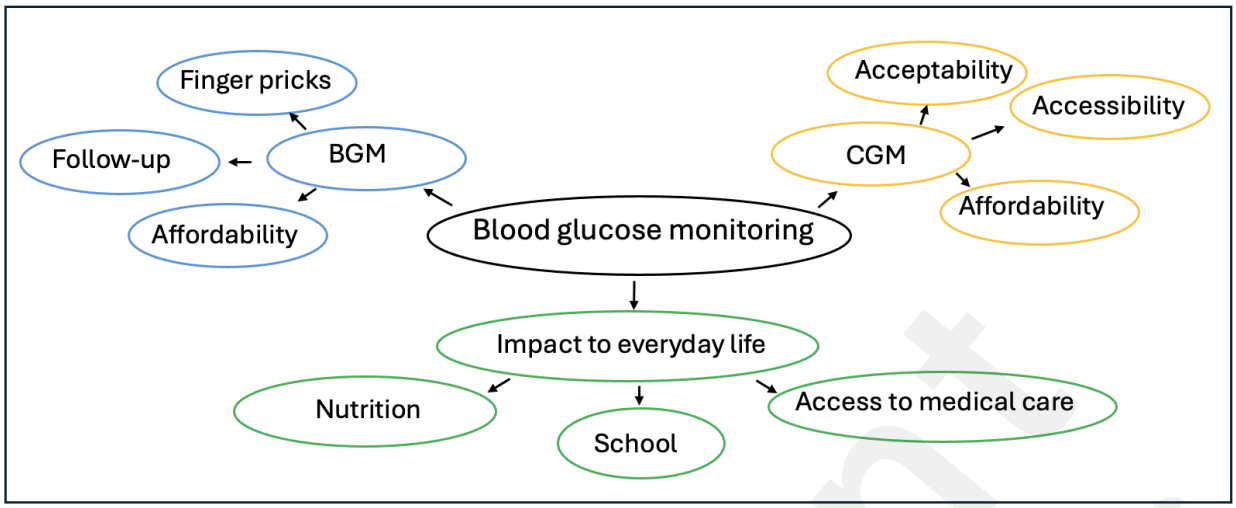
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Table 1 – Description of each country

	Kyrgyzstan	**Mali**	**Peru**	**Tanzania**
Geographical location	Central Asia	West Africa	South America	East Africa
Population, total (2023) [17]	7,100,800	23,293,698	34,352,719	67,438,106
Life-expectancy at birth (total years) (2022) [17]	72	59	73	67
Poverty headcount ratio at $2.15 a day (% of population) (2017) [17]	0.3	20.8	2.7	44.9
Diabetes-related health expenditure per person, USD (2021) [18]	231	189.6	1331.5	No data available
T1DM (0-19 y), in 1000s [18]	0.4	0.3	0.4	No data available
Regions included in assessment	Bishkek (capital)	Bamako (capital)	Lima (capital)	Dar es Salaam (main urban area)

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Figure 1 – Themes emerging from the interviews



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Table 2 – Summary of results related to BGMs and CGMs

Barrier/Challenge	Solution provided by BGM	Solution provided by CGM
Price	High depending on number of tests	Out of reach and unaffordable for many
Ease of use	Easy and familiarity to use	Needs further knowledge and education
Psychological stress for caregivers	Known use, confidence in equipment, high number of values reduces stress (limited by equipment and costs)	Need to become familiar with the equipment, education, monitoring of blood glucose trends to anticipate and reduce stress
Purchasing access	Normally everywhere	Not available formally in the countries included in this study
Technology access	Smartphones not needed, electricity no needed	Technology dependent
Glycemic control	Measurement-dependent, highly reliable instantaneous value	Continuous, including nights alarm system, remotely accessible data
Finger prick “trauma”	Each time	At installation, for calibration, mainly avoided
Aspects and activities of daily life	Provides a reliable measurement for adapting activities (nutrition, sports), requires dedicated time and handling, equipment to be carried with you, caregiver present (depending on the child's autonomy)	Allows continuous measurement of blood glucose levels in the event of variations (diet, sport), no special measuring effort or additional equipment required, remote control (practical for caregivers)
Self-education	Using glycemia’s values recorded in a notebook, no technology needed, requires many values	Better understanding of blood glucose trends, values may be incorrect and therefore requires knowledge of hypoglycemia’s symptoms
Medical support for care providers	Depends on notebooks, reliable values in principle (if not incorrectly noted), Bluetooth system on some BGMs	Useful support, data sharing easy
Confidence, acceptability	High confidence and acceptability	Need for more information, good confidence among CGM users, issue of visibility of device